

DEVA ANAND

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Education

MS, Computer Science , University of Massachusetts Amherst	2025–2027
<i>Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering</i>	
BE, Computer Science and Engineering , Vels Institute of Science, Technology and Advanced Studies	2019–2023

Professional Experience

TherAlign Health , Founding Software Engineer Intern <i>Remote</i>	Jul 2026 – Present
- Founding engineer building full-stack features (Next.js + Firebase/GCP) for a provider-facing SMART on FHIR clinical decision support app that surfaces covered, guideline-backed medication alternatives to prevent prior authorization. - Extending a Firebase Cloud Functions backend built from RxNorm normalization, formulary/prior-auth logic, and PubMed/guideline evidence, with Gemini constrained to synthesis while the backend deterministically controls eligibility and coverage so generative output stays grounded and verifiable.	
Culture & Morality Lab, UMass Amherst , Lab Assistant <i>Amherst, MA</i>	Jul 2026 – Present
- Sole engineer independently building the lab's CCR text-analysis platform (Python, FastAPI, sentence-transformers, React), productionizing a published NLP method into a self-serve full-stack web tool for researchers. - Built a YAML model registry and versioned construct library (99 constructs from 38 questionnaires); drove development with a Claude Code multi-agent setup (review agents, protective hooks, golden-file guards) and an every-session test discipline that grew the backend suite to 40 hermetic tests.	
Wysa , Backend Engineer <i>Bengaluru, India</i>	Jul 2023 – Jul 2025
- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for 8+ UK clinical clients, processing 10,000+ monthly triage submissions via Mayden iaptus REST integrations; contributed to \$340K+ in revenue. - Architected a full-stack ML training-data annotation platform (React, Node.js, MongoDB) used daily by the AI team, cutting manual annotation by 100+ hours/month and lifting labeling throughput by 60%. - Piloted a COO-sponsored GitHub Copilot productivity experiment across code migrations, tech-debt PRs, and security-finding triage, maintaining a versioned <code>copilot-instructions.md</code> conventions file to keep AI-generated code aligned with codebase patterns.	

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub Demo)	Python, FastAPI, OpenAI / Groq, SQLite
Built and deployed a multi-stage agentic web-discovery service turning open-ended queries into structured entity tables via typed retrieval planning, reranking, and deterministic extraction with LLM fallback and cell-level provenance; hardened with 150+ tests so AI output stays validated, not trusted blindly.	
cf-ai-research-scout – Edge-Native AI Assistant (GitHub)	TypeScript, Cloudflare Workers AI, Durable Objects, D1
Shipped an end-to-end edge-native AI agent on Cloudflare Workers that fetches web sources and streams multi-step research digests over WebSocket, with durable workflows and Durable Objects that survive process restarts and tool failures without an external database.	
Sonare – Offline Sign / Speech App (GitHub)	React, Electron, FastAPI, MediaPipe, whisper.cpp
Built a cross-platform desktop app under hackathon time pressure (Qualcomm Edge AI 2025) with a fully offline on-device pipeline (MediaPipe hand tracking, whisper.cpp ASR, local TTS) and async queues on a FastAPI + Electron/React stack.	

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & Databases: Python, TypeScript, JavaScript, C++, Java, SQL, MongoDB, PostgreSQL, SQLite
Full-Stack: Next.js, React, Node.js, Express, FastAPI, REST APIs, SDKs, Docker, Git, GitHub Actions
AI-Assisted Dev: Claude Code, Cursor, GitHub Copilot, OpenAI Codex, ChatGPT
ML & AI: PyTorch, Hugging Face Transformers, sentence-transformers, LLM agents, RAG, MCP, OpenAI / Gemini APIs
Cloud & DevOps: AWS, GCP, Azure, Cloudflare Workers, Firebase, Linux

Open Source & Community

- Built **Underdogs Fitness** (Website), a production MERN platform with Stripe payments; live **3+ years**. - Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.